power

File: power.kicad_sch

mcu

File: mcu.kicad_sch

sensing

File: sensing.kicad_sch

vio

File: vio.kicad_sch

translators

File: translators.kicad_sch

connectors

File: connectors.kicad_sch

OOTY rev A — hierarchical schematic generated by scripts/build_schematic.py

Specification: SPEC.md — design notes and calculations: NOTES.md

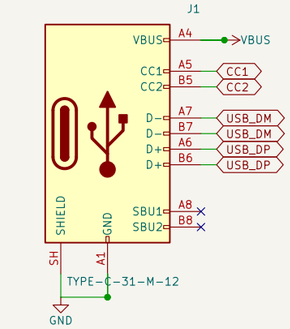
OOTY

Sheet: /
File: OOTY.kicad_sch

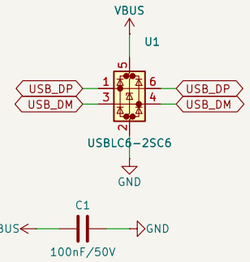
Title: OOTY — USB Hardware-in-the-Loop board

Size: A4	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/7

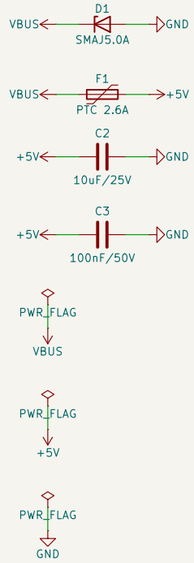
USB-C input



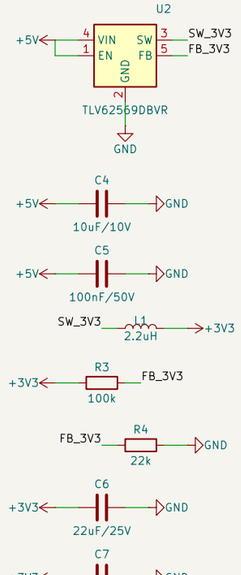




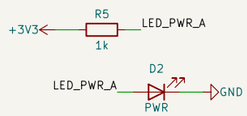
VBUS protection



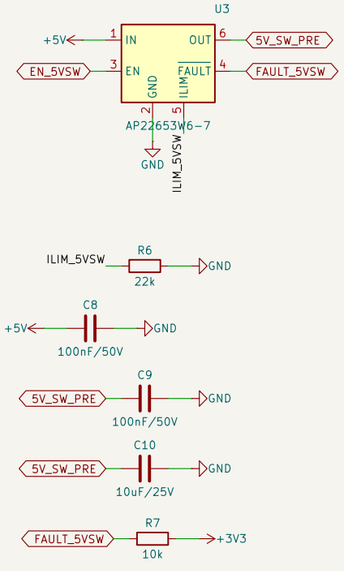
3V3 buck (TLV62569)



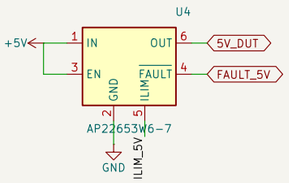
Power OK LED



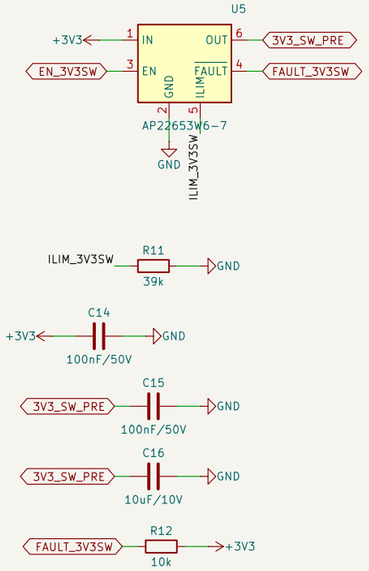
Switch 5V_SW (switched, 1 A)



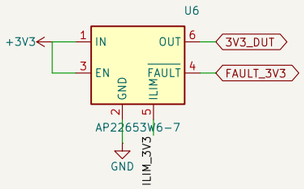
Switch 5V fixed (0.5 A)



Switch 3V3_SW (switched, 0.5 A)



Switch 3V3 fixed (0.3 A)



Rail test points

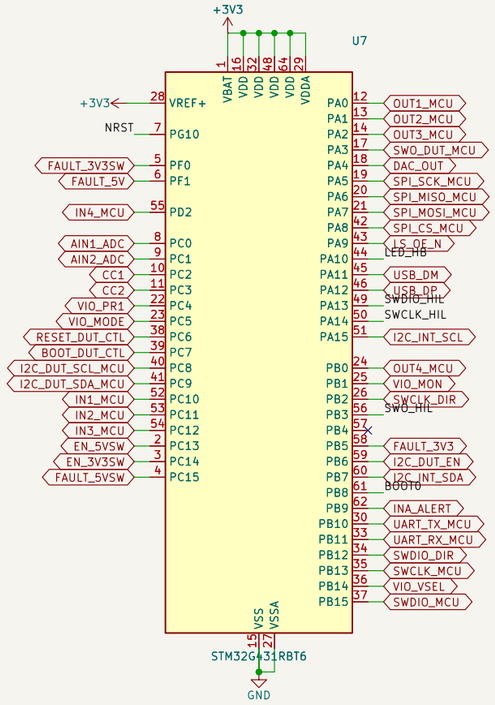


AP22653 ILIM (DS41186 p.10, VIN = 5 V): 22k -> 1.16 A typ, 39k -> 0.64 A typ, 68k -> 0.35 A typ. See NOTES.md.

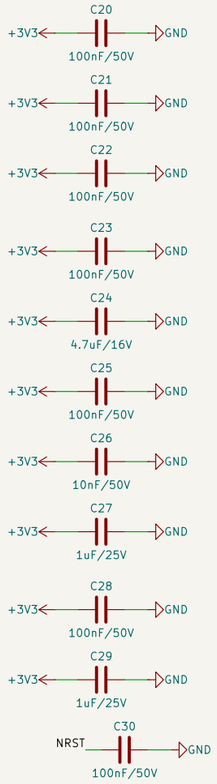
Buck: Vout = 0.6 V x (1 + 100k/22k) = 3.327 V.

OOTY		
Sheet: /power/		
File: power.kicad_sch		
Title: Power: USB-C, 3V3 buck, DUT rails		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/7

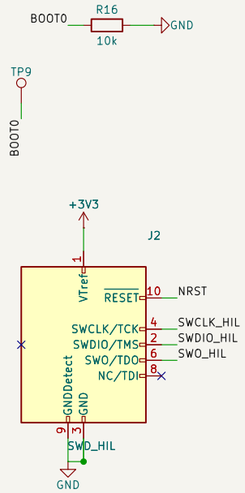
STM32G431RBT6



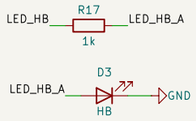
Decoupling (DS12589 5.1.6)



Boot / HIL debug (SWD)



Heartbeat LED

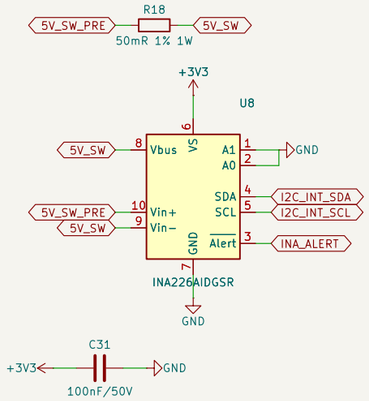


USB FS runs crystal-less (HSI48 + CRS). BOOT0 test pad: short to 3V3 for the system bootloader.

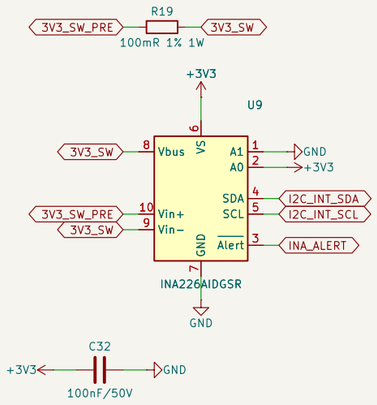
PB4/PB6 carry UCPD dead-battery pull-downs at reset: PB4 unused, PB6 drives I2C_DUT_EN (low = disabled, safe default).

OOTY		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: MCU STM32G431RBT6		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/7

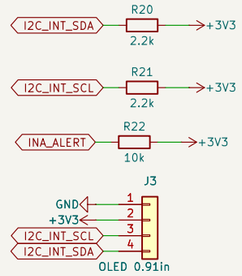
Current monitor 5V_SW



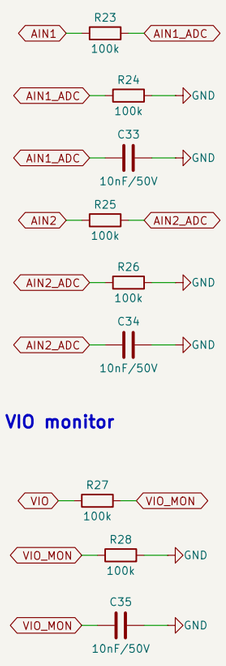
Current monitor 3V3_SW



Internal I2C + OLED

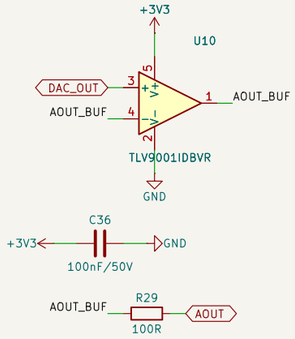


Analog inputs (0–5.5 V)



VIO monitor

DAC output buffer

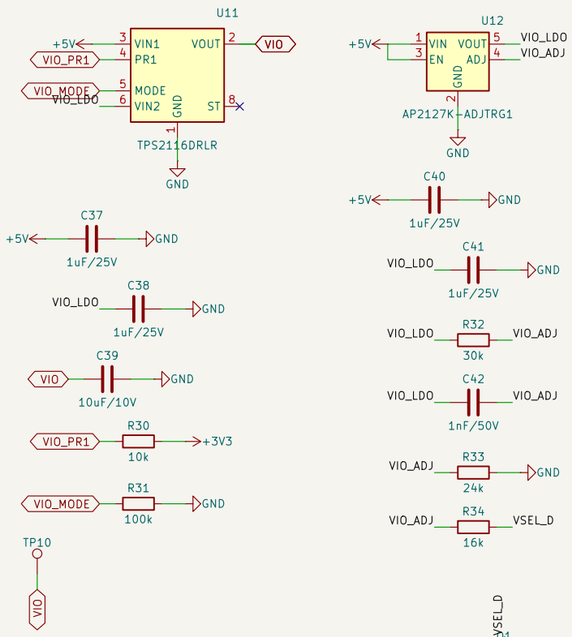


INA226 addresses: 5V_SW 0x40 (A1=A0=GND), 3V3_SW 0x41 (A0=VS). OLED 0x3C. INA228 is a drop-in replacement (same pinout).

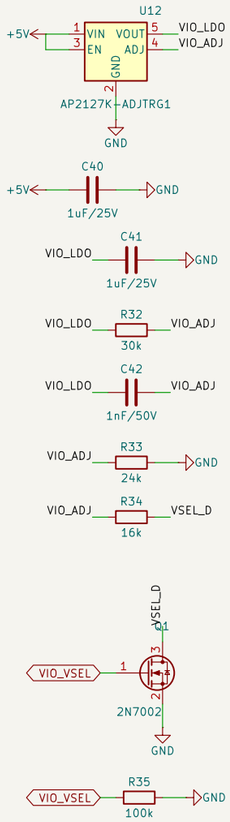
Shunts: Kelvin connection to IN+/IN− required at layout.

OOTY		
Sheet: /sensing/		
File: sensing.kicad_sch		
Title: Current sensing, OLED, analog I/O		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/7

VIO selection (TPS2116 mux)



VIO LDO 1.8 / 3.3 V (AP2127K-ADJ)

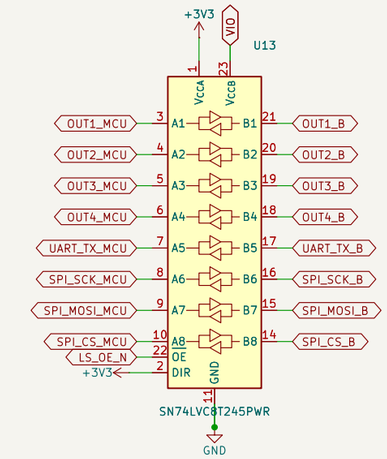


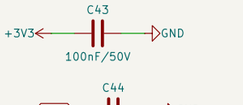
TPS2116: MODE=1 PR1=1 -> 5V_SYS ; MODE=1 PR1=0 -> LDO ; MODE=0 PR1=1 -> off (external VIO from J_PWR/J_DBG). MODE=0 PR1=0 is forbidden (selects 5 V).

LDO: Vout = 0.8 x (1 + 30k/24k) = 1.80 V ; VIO_VSEL high adds 16k in parallel -> 3.30 V.

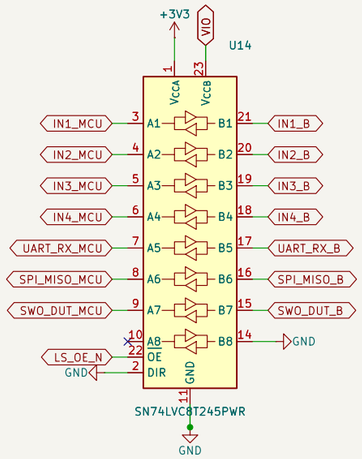
OOTY		
Sheet: /vio/		
File: vio.kicad_sch		
Title: VIO: 1.8 / 3.3 / 5 V or external		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/7

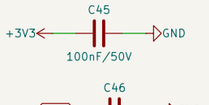
Outputs MCU -> DUT (SN74LVC8T245, DIR = A to B)



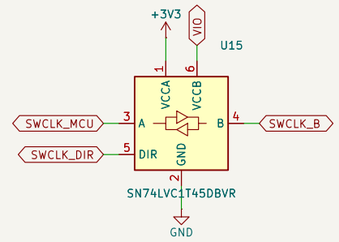


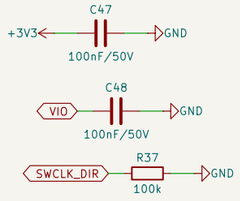
Inputs DUT -> MCU (SN74LVC8T245, DIR = B to A)

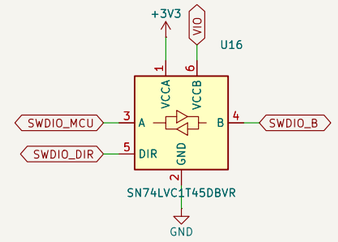


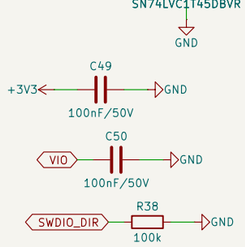


SWD probe (SN74LVC1T45)

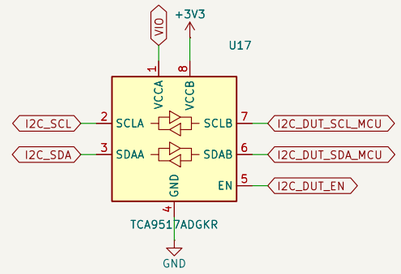


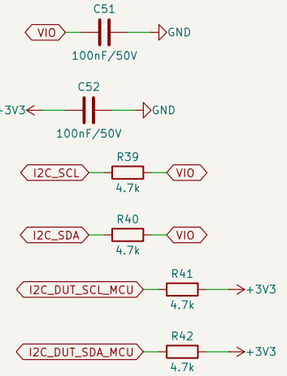




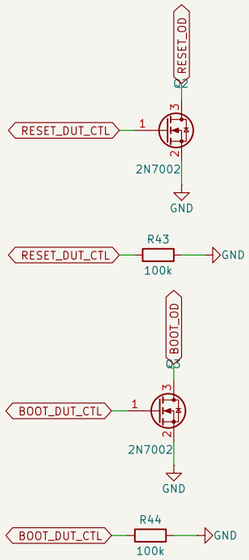


DUT I2C buffer (TCA9517A, A = DUT, B = MCU)





RESET / BOOT open-drain



All translators are Hi-Z until firmware drives LS_OE_N low / DIR high / I2C_DUT_EN high.

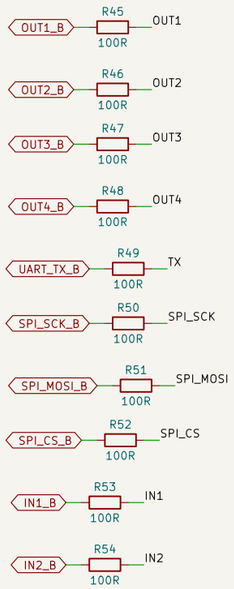
OOTY

Sheet: /translators/
File: translators.kicad_sch

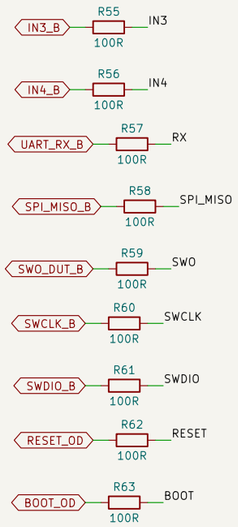
Title: Level translators 3V3 <-> VIO

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 6/7

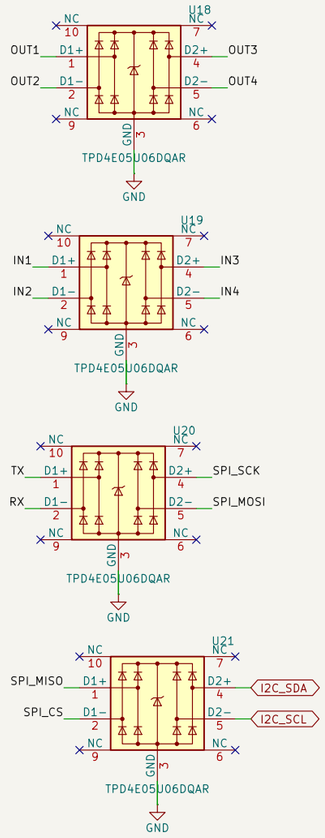
Series resistors



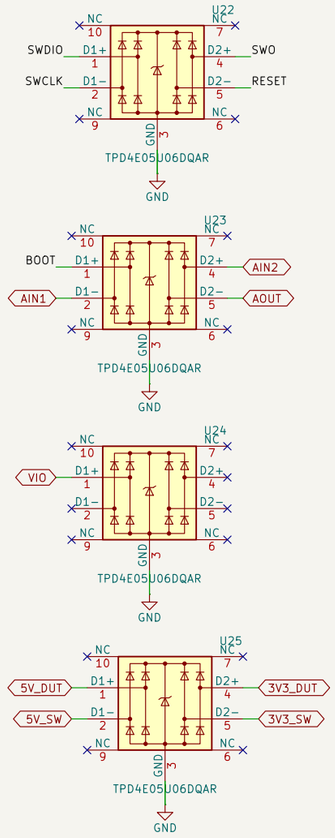
Series resistors (cont.)



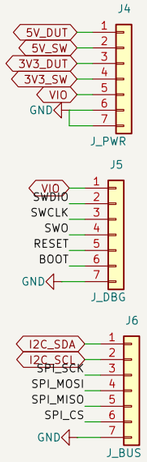
ESD (TPD4E05U06)



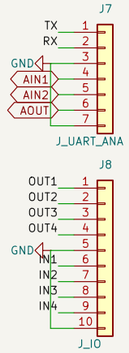
ESD (cont.)



DUT headers (2.54 mm)



DUT headers (cont.)



Mounting holes



Signal path: header -> TVS -> 100R series -> translator / circuit.

OOTY		
Sheet: /connectors/		
File: connectors.kicad_sch		
Title: DUT connectors and ESD		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 7/7